Reg. No. [ ] [14AE45]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2016

AEROSPACE STRUCTURES-I

Time: 3 Hours Maximum Marks:70

Instructions:

1. Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART-A

1. a. Explain the following: 05

   i. Law of Superposition

   ii. Statically determinate and indeterminate structures

   iii. Degree of Indeterminacy

   iv. Linear systems

   v. Non-linear Systems

   b. Classify the beams as determinate or indeterminate and if indeterminate report the degree of indeterminacy. Loads on the beams are known and can act anywhere. 04

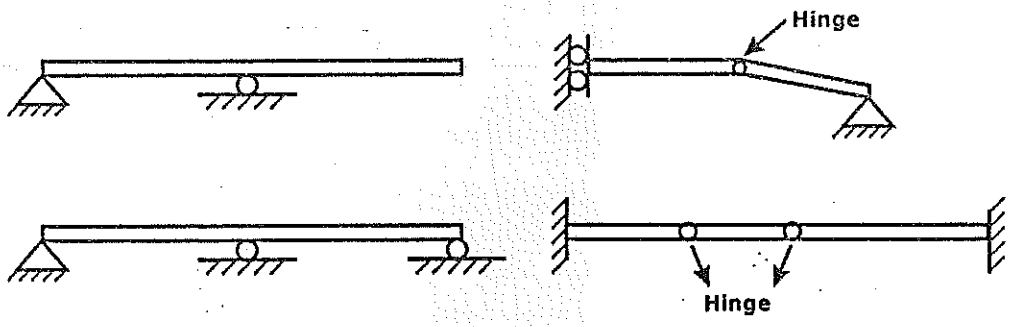

Figure 1(b)

   c. Find the support reactions at the point A in the figure 1(c). 05

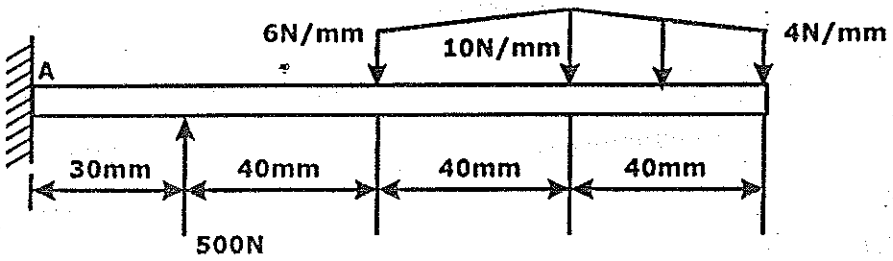

Figure 1(C)

2. a. Explain with neat diagrams the method of joints and method of section used in the analysis of truss system. 06

b. For the truss shown in figure 2(b), find the reactions and forces in the members DE, CE and BC (Use method of Sections). 08

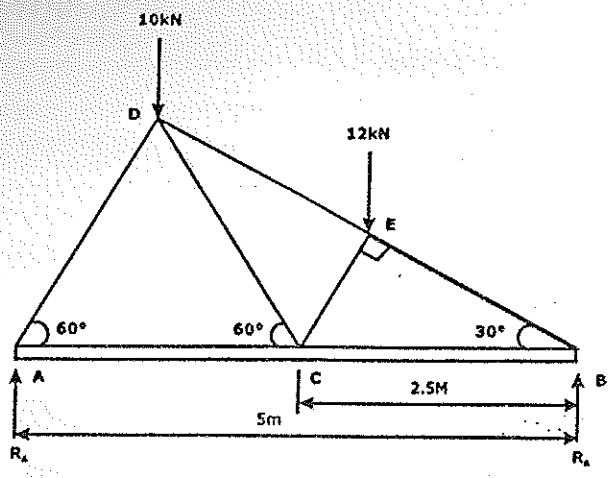

Figure 2(b)

3. a. A rectangular column of width 200mm and thickness 150mm shown in figure 3(a), carries a point load of 240kN at an eccentricity of 10mm. Determine the maximum and minimum stresses on the section. 07

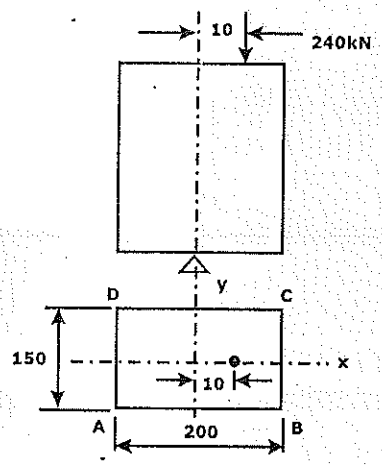

Figure 3(a)

b. In the above problem **(3.a)**, if the minimum stress on the section is zero, then find the eccentricity of the point load of 240kN, acting on the column. Also calculate the corresponding maximum stress on the section. 07

4. a. What do you mean by shear stress in beams? Derive the expression for the shear stress at any point in a circular section of a beam, which is subjected to a maximum shear force F. 10

b. A rectangular beam of 100mm wide and 250mm deep is subjected to a maximum shear force of 75kN. Determine : 04

i. Average Shear Stress

ii. Maximum Shear Stress

PART-B

5. a. What is shear centre? Show that the location of shear centre for a channel section is given by the expression **14**

$$e = \frac{h^2 \times t_1 \times b^2}{4I_{xx}}$$

Where

h = distance between the center lines of the flanges

t_1 = Thickness of the flanges of I-Section

b = Longer length of I-Section from vertical center line of the web

I_{xx}=Moment of Inertia of I-Section about axis of symmetry (X-X axis)

E = Horizontal distance of applied vertical downward shear force from center of web.

6. a. Explain elastic strain energy. Write the strain energy expression under different loads: Uniaxial stress system, Simple shear. **?**

b. For the beam shown in the figure 6(b), hinged at C and a hinge support at B, determine the reactions at C and B (Use Virtual work Principle). **?**

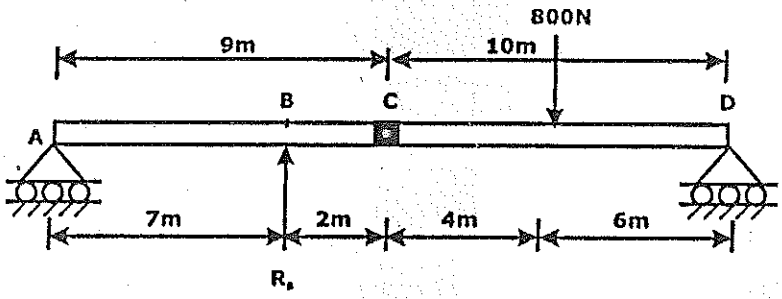

Figure 6(b)

7. a. State Maxwell's Reciprocal theorem for: **04**

i. Deflection and Unit load

ii. Moments and Rotations.

b. Determine the deflection and rotation at the free end of the cantilever beam shown in figure 7(b). Employ unit load method. **10**

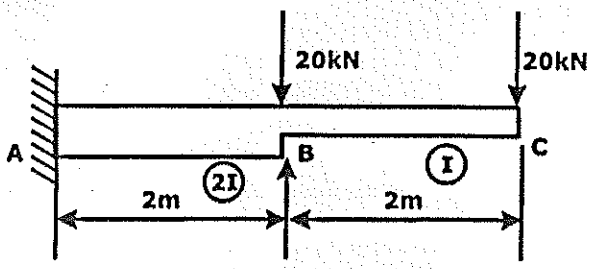

Figure 7(b)